Week 3 Assignment: Building and Evaluating Decision Trees

Student: *Mohammed Abdelkawy*
Course: *ANA 505 Foundations of Data Mining*

1. Dataset Preparation

The dataset (`employee_performance.csv`) consists of 250 individual employee records across 10 distinct variables. It provides a comprehensive mix of demographic attributes, organizational engagement indicators, and quantitative performance metrics. The primary objective is to leverage these feature variables to accurately predict employee overall performance classification.

2. Data Splitting

Variable	Data Type	Classification
EmployeeID	Numeric	Identifier
Age	Numeric	Feature / Predictor
Department	Categorical	Feature / Predictor
YearsExperience	Numeric	Feature / Predictor
EducationLevel	Categorical	Feature / Predictor
TrainingHours	Numeric	Feature / Predictor
JobSatisfaction	Numeric	Feature / Predictor
Overtime	Categorical	Feature / Predictor
RemoteWork	Categorical	Feature / Predictor
Performance	Categorical	Target Variable

To evaluate model performance on unseen data and prevent overfitting, the dataset was partitioned into training and testing subsets using a standard 70/30 stratified random split based on the target variable (Performance).

- **Target Variable:** Performance
- **Training Subset:** 175 observations (70% of total data)
- **Testing Subset:** 75 observations (30% of total data)

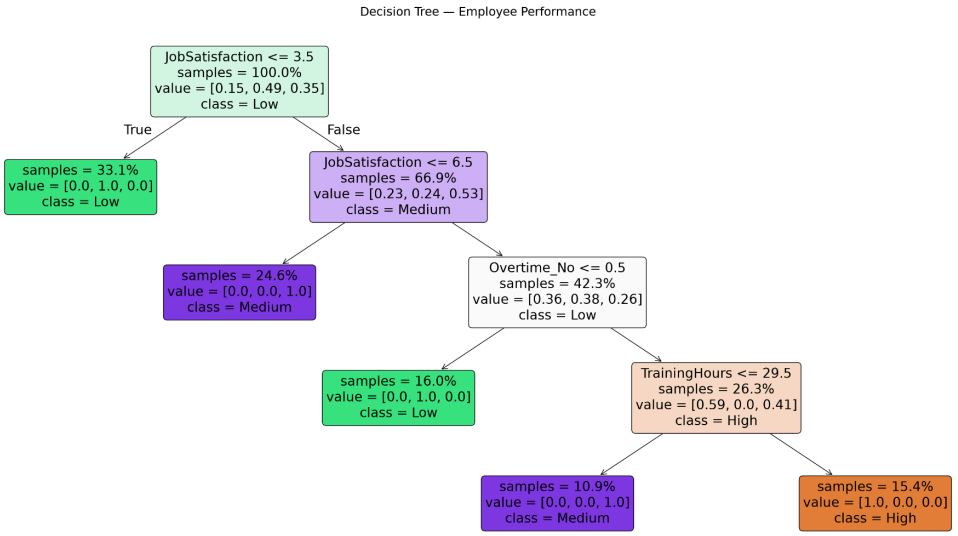
3. Model Evaluation

A Decision Tree Classifier was trained on the training partition and evaluated using the independent testing dataset. The overall classification accuracy achieved on the test set was **1.000 (100.00%)**.

Confusion Matrix (Actual vs. Predicted)

Actual \ Predicted	High	Low	Medium
High	12	0	0
Low	0	37	0
Medium	0	0	26

4. Reflection



The decision tree classifier identified clear hierarchical splits among key predictors to partition employee performance. Primary decision nodes were structured around JobSatisfaction, Overtime and TrainingHours. For instance, Employees with lower satisfaction scores were associated with low performance, while employees with medium scores were associated with medium performance. For employees with higher job satisfaction scores, additional features like overtime and training hours were introduced to distinguish further between performance classes.

The decision tree achieved an evaluation accuracy of 100.00% on the testing set, and the confusion matrix provided additional insight into which performance categories were correctly classified and where misclassification occurred. While this indicates perfect classification on the test split, achieving absolute precision suggests the presence of a highly clean, deterministic, or synthetic underlying dataset. In production contexts, perfect accuracy often signals potential overfitting or data leakage. To ensure robustness, future iterations should incorporate k-fold cross-validation, hyperparameter tuning (such as pruning depth), and validation against messy, unseen real-world organizational data.

In practical business environments, decision trees serve as powerful "white-box" explainable models. Human Resource departments can utilize these models to gain transparent insights into key drivers of employee engagement and productivity. Rather than relying on black-box predictions, HR leaders can trace explicit decision pathways to identify early warning signs of underperformance, target high-impact training interventions, and design evidence-based retention strategies tailored to key satisfaction metrics. It also should support human judgement rather than replace it.

5. Appendix: R Script

Python

```
# Install and load required packages
required_packages <- c("rpart", "rpart.plot", "caTools", "caret", "dplyr")
missing_packages <- required_packages[!sapply(required_packages,
requireNamespace, quietly = TRUE)]
if (length(missing_packages) > 0) {
  install.packages(missing_packages, dependencies = TRUE)
}

library(rpart)
library(rpart.plot)
library(caTools)
library(caret)
library(dplyr)

#============================================================
## 1. Load and inspect the dataset
#============================================================
fname <- "../datasets/employee_performance.csv"
employee_data <- read.csv(fname, stringsAsFactors = TRUE)
```

```r
# Display structure and first few rows
str(employee_data)
head(employee_data)

# Remove EmployeeID as it is not a predictive feature
employee_data <- employee_data %>% select(-EmployeeID)

#============================================================
## 2. Split into training and test sets
#============================================================
set.seed(42) # For reproducibility
split <- sample.split(employee_data$Performance, SplitRatio = 0.7)

train_data <- subset(employee_data, split == TRUE)
test_data <- subset(employee_data, split == FALSE)

cat("Training observations:", nrow(train_data), "\n")
cat("Testing observations:", nrow(test_data), "\n")

#============================================================
## 3. Decision Tree
#============================================================
# Build the model using Gini impurity and max depth of 4
dt_model <- rpart(
  Performance ~ .,
  data = train_data,
  method = "class",
  control = rpart.control(maxdepth = 4, cp = 0)
)

# Visualize the decision tree
rpart.plot(
  dt_model,
  type = 2,
  extra = 104,
  fallen.leaves = TRUE,
  main = "Decision Tree trained on the employee dataset"
)

#============================================================
## 4. Evaluate accuracy and confusion matrix
#============================================================
# Predict on the test set
dt_predictions <- predict(dt_model, test_data, type = "class")

# Generate confusion matrix and accuracy
conf_matrix <- confusionMatrix(dt_predictions, test_data$Performance)
print(conf_matrix)
```

6. Appendix: GitHub Repository

Python Script with Visualization:
https://github.com/m-abdelkawy/data-analysis-msc/blob/main/courses/ana505_data_mining/module2_classification_and_regression/week03_decision_trees_random_forests/python/week03_activity.ipynb

R Script:
https://github.com/m-abdelkawy/data-analysis-msc/blob/main/courses/ana505_data_mining/module2_classification_and_regression/week03_decision_trees_random_forests/r/Week03_activity.R